Pronob Sarker

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RESEARCH INTERESTS

Machine Learning, Robust and Trustworthy AI, Large Language Models, AI agents,Human Computer Interaction, Computer Vision, AI Reasoning and Evaluation, Optimization

EDUCATION

Gettysburg College, Gettysburg, Pennsylvania, USA — Aug 2022 – Dec 2026
Bachelor of Science in Computer Science
Minors: Data Science and Mathematics — GPA: 3.71/4.00

RESEARCH EXPERIENCE

Machine Learning Research Assistant — Spring 2025
Gettysburg College & Rice Fruit Company
Faculty Advisor: Dr. Kim Spayd

- Conducted a comparative study of KNN, SVM, Random Forest, and ResNet-50 for fine-grained apple image classification using the Fruit-360 dataset.
- Implemented and evaluated models using accuracy, precision, recall, macro-F1, and confusion matrices, with ResNet-50 achieving a macro-F1 score of 0.99.
- Analyzed misclassification patterns among visually similar apple classes to investigate model generalization and practical deployment trade-offs.

Graph Theory Research — Spring 2024
Gettysburg College
Faculty Advisor: Prof. Nathan Shank

- Defined two parametric graph families, Firecracker and Tollbooth graphs, using formal vertex-edge constructions.
- Proved closed-form results involving domination number, Eulerian properties, graph diameter, and independence number.
- Developed the research project *"Firecrackers and Tollbooth Graphs: Properties of Novel Graph Structures."*

Undergraduate Research Assistant – Game AI — Summer 2023
Gettysburg College
Faculty Advisor: Prof. Todd W. Neller

- Investigated Monte Carlo sampling and stochastic local-search approaches for generating and evaluating combinatorial puzzles.
- Optimized algorithms used to generate and rank more than 1,000,000 solvable puzzle instances.
- Conducted empirical analysis of generated puzzles to evaluate diversity and distinguish levels of puzzle difficulty.

PUBLICATIONS & RESEARCH OUTPUTS

Research Manuscript / Project

- *Firecrackers and Tollbooth Graphs: Properties of Novel Graph Structures.* Undergraduate graph theory research project, Gettysburg College, 2024.

SELECTED AI & TECHNICAL PROJECTS

Unofficial Guide – Retrieval-Augmented Generation System — 2026

- Developed a retrieval-augmented generation application for answering questions over institutional documents and guides.
- Built a document-processing and retrieval pipeline using pdfplumber, ChromaDB, and sentence-transformer embeddings.
- Integrated LLM inference and an interactive Gradio interface for grounded question answering over retrieved context.

FitFindr – AI Agent with Tool Use — 2026

- Developed an AI-agent application using LLM function calling to dynamically select tools and generate personalized responses.
- Implemented structured tool execution and an interactive Gradio interface for agent-based workflows.

WhatTheHealth – AI-Powered Healthcare Decision Support System — HackUMBC 2024

- Built an end-to-end AI application integrating natural language processing and supervised machine learning for risk prediction from free-text symptom descriptions.
- Developed an NLP pipeline using spaCy to extract structured features from unstructured text and trained machine-learning models on curated medical datasets.
- Integrated Python inference services with a Next.js frontend through REST APIs and voice-to-text functionality.

PROFESSIONAL EXPERIENCE

Software Engineering Intern Summer 2026
Samskrita Bharati USA

- Contributed to the development of a Django-based volunteer management platform for coordinating programs, volunteers, and organizational workflows.
- Implemented role-based access control, task-management workflows, volunteer-facing interfaces, and program coordination features.
- Developed workflows for bulk task creation, volunteer acknowledgement, email notifications, search/filtering, and administrative dashboards.

Software Engineering Intern – User Experience Summer 2025
Develop for Good

- Designed Python-based LLM-assisted pipelines for large-scale cloud content migration, reducing deployment latency from approximately three weeks to one week.
- Built automated data-ingestion and preprocessing workflows using Python, BeautifulSoup, and REST APIs for more than 5,000 heterogeneous assets.
- Led peer code reviews and validation checks, identifying more than 20 pre-release defects and improving system reliability and reproducibility.

Software Engineering Intern – Admission Infobot Summer 2022
Nextive Solutions

- Developed a Messenger-based college admission chatbot serving more than 5,000 students and automating frequently asked admission questions.
- Implemented interactive workflows through the ManyChat API, reducing repetitive inquiries and accelerating access to admission information.

TEACHING & MENTORING

Teaching Assistant – Machine Learning 2026
Gettysburg College

- Support students with core machine-learning concepts, programming assignments, model implementation, and debugging.
- Assist with course activities and provide technical guidance on machine-learning workflows and experimentation.

AWARDS, FELLOWSHIPS & GRANTS

Emory University Laney Graduate School Summer Opportunity for Academic Research (LGS-SOAR) 2026
Selected participant in the graduate-school research and scholarly development program.

NSF-Funded Polymath Jr. Summer Research Program 2025
Selected participant in the collaborative undergraduate mathematics research program.

Davis Projects for Peace Grant 2023
Awarded a competitive $10,000 grant to design and lead *Bridging the Digital Divide* in Bogura, Bangladesh.

X-SIG Summer Research Fellowship 2023
Selected for Gettysburg College's summer research fellowship program.

Presidential Scholarship 2022
Awarded by Gettysburg College in recognition of academic achievement.

Dean's List Multiple Semesters
Recognized for academic excellence.

Stellar Student Employee Award
Recognized for outstanding contributions as a student employee.

PROFESSIONAL SERVICE & LEADERSHIP

President – Google Developer Student Club 2025
Gettysburg College Chapter

- Organized G-Hacks, a student mini-hackathon with more than 30 participants.
- Organized recurring technical and algorithmic problem-solving activities for students.

Project Lead – Bridging the Digital Divide Jun 2023 – Aug 2023
Davis Projects for Peace, Bogura, Bangladesh

- Led a $10,000-funded digital-literacy and educational-access initiative serving more than 500 students.
- Secured an additional $2,000 in funding to expand the initiative to additional communities.

SELECTED COURSEWORK

AI, ML & Mathematics
Machine Learning
Deep Learning
Linear Algebra
Multivariable Calculus
Differential Equations
Mathematical Modeling
Applied Research in Mathematics

Statistics, Theory & Systems
Probability & Statistics
Applied Statistics
Theory of Computation
Data Structures & Algorithms
Computer Organization
Computer Networks
Database Management Systems

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, SQL, JavaScript
- **Machine Learning / AI:** PyTorch, TensorFlow, scikit-learn, spaCy, Large Language Models, Retrieval-Augmented Generation
- **Development & Research Tools:** Git, AWS, REST APIs, LaTeX, MATLAB, Mathematica, ChromaDB, Gradio
- **Methods:** Machine Learning, Deep Learning, Computer Vision, Statistical Analysis, Algorithm Design, Monte Carlo Methods

REFERENCES

Prof. Todd W. Neller
Professor of Computer Science, Gettysburg College
Gettysburg, Pennsylvania, USA
E-mail: `[confirm email]`

Prof. Eatai Roth
Department of Computer Science, Gettysburg College
Gettysburg, Pennsylvania, USA
E-mail: `[confirm email]`

Prof. Clifton Presser
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